

máme $A = \begin{pmatrix} 1 & 1 & 1 & -2 \\ 1 & 1 & -2 & 1 \\ 1 & -2 & 1 & 1 \end{pmatrix}$

hledáme singulární rozklad, tedy $A = UDV^T$, kde D bude diagonální typu 3×4 , V bude 4×4 a U 3×3

$$A^T = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -2 \\ 1 & -2 & 1 \\ -2 & 1 & 1 \end{pmatrix}$$

$$A^T A = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 6 & -3 & -3 \\ 0 & -3 & 6 & -3 \\ 0 & -3 & -3 & 6 \end{pmatrix} \rightarrow \text{vl. č. } \lambda_1 = 9 \quad v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}$$

(wolfram mathematica)

$$\lambda_2 = 9 \quad v_2 = \begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \end{pmatrix} \quad \sigma_2 = 3$$

násobí hodnotu $\rightarrow$ upravíme matice pro $(A^T A - 9I_4): \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix}$

$$D = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix}$$

$$\lambda_3 = 3 \quad v_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \sigma_3 = \sqrt{3}$$

$$\lambda_4 = 0 \quad v_4 = \frac{1}{\sqrt{3}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$v_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 0 \\ -1 \\ 2 \\ -1 \end{pmatrix}$$

$$\hookrightarrow V = \begin{pmatrix} 0 & 0 & 1 & 0 \\ -1/\sqrt{2} & -1/\sqrt{6} & 0 & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 0 & 1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & 0 & 1/\sqrt{3} \end{pmatrix}$$

obtývá tedy matice U bu bude typu 3×3 , navíc tedy $u_i = \frac{1}{\sigma_i} A v_i$ (pro singulární čísla)

$$u_1 = \frac{1}{3} A \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{3\sqrt{2}} \begin{pmatrix} -1 \\ 1 \\ 1 \\ -2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \\ 1 \\ -2 \end{pmatrix}$$

$$u_2 = \frac{1}{3} A \frac{1}{\sqrt{6}} \begin{pmatrix} 0 \\ -1 \\ 2 \\ -1 \end{pmatrix} = \frac{1}{3\sqrt{6}} \begin{pmatrix} -1 \\ 1 \\ 2 \\ -3 \end{pmatrix} = \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ 1 \\ 2 \\ -3 \end{pmatrix}$$

$$u_3 = \frac{1}{\sqrt{3}} A \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \\ -2 \end{pmatrix}$$

$$\rightarrow U^T = \begin{pmatrix} -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{pmatrix}$$

$$\hookrightarrow A = U D V^T = \begin{pmatrix} -1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \\ 0 & -2/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix} \begin{pmatrix} 0 & -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & -1/\sqrt{6} & 2/\sqrt{6} & -1/\sqrt{6} \\ 1 & 0 & 0 & 0 \\ 0 & 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{pmatrix}$$

$$A = U D V^T$$

$$A^T = (U D V^T)^T = (V^T)^T D^T U^T = V D^T U^T$$

$$A^T = \begin{pmatrix} 0 & 0 & 1 & 0 \\ -1/\sqrt{2} & -1/\sqrt{6} & 0 & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 0 & 1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & 0 & 1/\sqrt{3} \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{pmatrix}$$

$$A A^T = U D V^T V D^T U^T = U D D^T U^T = \begin{pmatrix} -1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \\ 0 & -2/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \end{pmatrix} \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{pmatrix}$$

$V^T V = I_4$ (matice V je ortogonální)

$$A^T A = V D^T U^T U D V^T = V D^T D V^T =$$

$$\downarrow$$

$$U^T U = I_4 \text{ (matrice } U \text{ je ortogonalna)}$$

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ -1/\sqrt{2} & -1/\sqrt{6} & 0 & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 0 & 1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & 0 & 1/\sqrt{3} \end{pmatrix} \begin{pmatrix} 9 & 0 & 0 & 0 \\ 0 & 9 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & -1/\sqrt{6} & 2/\sqrt{6} & -1/\sqrt{6} \\ 1 & 0 & 0 & 0 \\ 0 & 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{pmatrix}$$